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DATA 205 - 33155
Maryland Insurance Administration

Analyzing Maryland Insurance Appeals and Grievances

Introduction:
In 1929, the first healthcare insurance plan was created, known as the Baylor Plan. This plan helped teachers by allowing them to pay a monthly fee of 50 cents in exchange for twenty-one days of hospital care. Since then, many insurance plans have been created with the intention of helping people pay for their medical services. However, it seems that these goals are not always being fully accomplished. Today, it is estimated that Americans owe between $194 billion and $220 billion in medical debt, which shows that affordability is still a major issue in the healthcare system.
The Maryland Insurance Administration (MIA) was created in 1993 to regulate insurance companies, protect Maryland consumers, and investigate complaints regarding insurance coverage.

Data Overview:
The data used for this project comes from the Appeals and Grievances Reports that are publicly available on the Maryland Insurance Administration website. I used data from 2021 through 2024, where each year has its own report. These reports contain detailed information about adverse decisions, grievances, and complaints.

Adverse decisions occur when an insurance claim is denied. If a consumer disagrees with that decision, they can file a grievance, which is an appeal of the denied claim. If the grievance is denied, the consumer can file a complaint with MIA. At that stage, the case may be reviewed by an Independent Review Organization, which provides an unbiased recommendation to MIA on how to proceed.

The datasets I used came from tables from Appendix 1, Appendix 2, and Appendix 8 of the reports. These included information such as insurer names, total counts, percentages, and breakdowns by service categories such as pharmacy, inpatient hospital, dental, physician services, and more.

Project Goals:
The goals of this project were to analyze trends in adverse decisions and grievances over time and to identify key patterns. Another goal was to identify patterns and trends in complaint outcomes and decisions. These goals are important because they help create a clear data story that can assist stakeholders in better understanding how insurance decisions are made and in identifying issues that need to be addressed. Possibly leading to policy changes that will ultimately help reduce the number of claim rejections for Maryland residents.

Tools:
The tools that were used in this project were Adobe Acrobat, Excel, R-Studio, and PowerBi.

Summary of data cleaning:
The data ingestion process for this project was complex and required multiple steps. Which includes downloading PDFs and extracting data into Excel files using Adobe Acrobat and Excel.
The full details of the data cleaning and ingestion process are available in my GitHub repository.
.https://github.com/KennyNguyen415/DATA-205-Spring-2026-/blob/main/Data Ingestion_Cleaning.pdf

Basic descriptive statistics:
For my exploratory data analysis, On R-Studio I created sixteen visualizations to better understand the dataset and answer key research questions which can be accessed in my GitHub repository. https://github.com/KennyNguyen415/DATA-205-Spring-2026-/blob/main/DATA_EDA.ipynb

One of the main questions I explored was "How the number of adverse decisions has changed over time?".

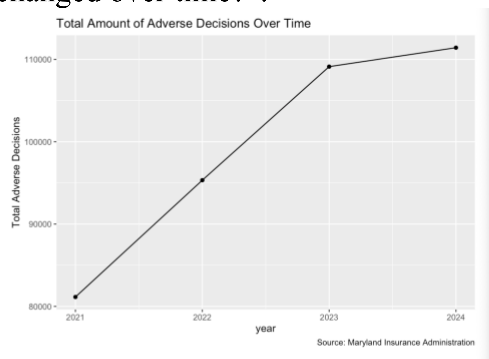

Figure 1. Line chart of total adverse decisions by year (2021–2024)

As shown in Figure 1, adverse decisions increased from approximately 81,143 in 2021 to 95,327 in 2022, then to 109,123 in 2023, and finally to 111,426 in 2024. This shows a clear upward trend, although the rate of increase slowed significantly between 2023 and 2024.
This trend is important because an increase in adverse decisions means that more claims are being denied, which can lead to higher out-of-pocket costs for consumers. For individuals living paycheck to paycheck, this can create serious financial stress. While speaking with Louis Buttler, the Director of the Appeals and Grievances Unit, he mentioned that the number of adverse decisions is unlikely to decrease, but efforts can be made to ensure that the increases are not as drastic over time.

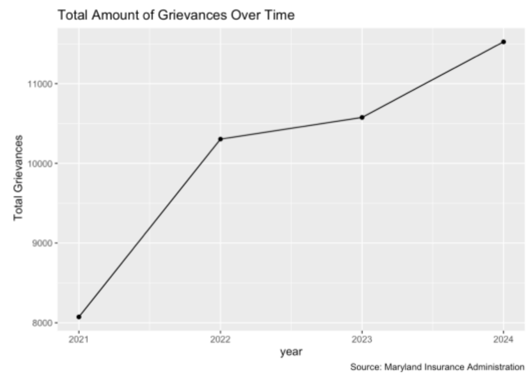

Figure 2. Line chart of total grievances by year (2021–2024

As shown in Figure 2, which is a line chart of grievances, grievances increased from 8,073 in 2021 to 10,305 in 2022, then slightly to 10,577 in 2023, and reached 11,526 in 2024. Like adverse decisions, grievances show an overall upward trend. However, this is not necessarily a negative outcome, as it may indicate that consumers are aware of their rights and are willing to appeal decisions.

Final data product:

My final data products were produced using PowerBi interactive dashboards. Since this project was completed with a sponsor, the dashboards were designed based on specific requests from Maryland Insurance Commissioner Marie Grant. One of the main requirements was to focus on the top five major medical companies and their associated entities, and to create dashboards that clearly show trends over time.

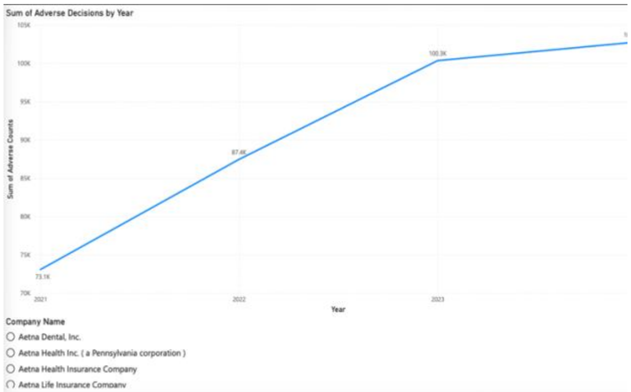

Figure 3. Power BI dashboard showing overall trends for adverse decisions

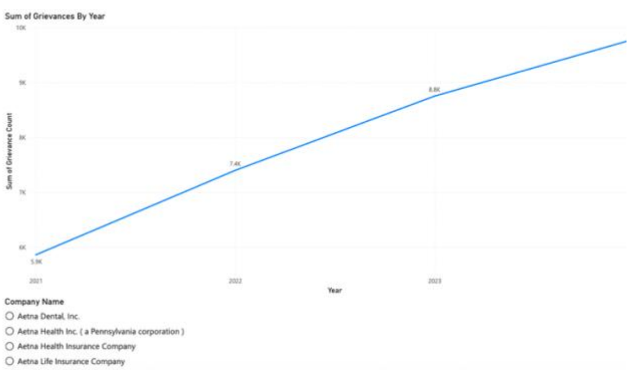

Figure 4. Power BI dashboard showing overall trends for grievances

Figures 3 and 4 show the overall trends for adverse decisions and grievances from 2021 to 2024. Commissioner Marie Grant requested a dashboard that presents these trends over time, and the data tells a consistent story of growth followed by stabilization. Adverse decisions increase sharply from 2021 through 2023, approaching the 100,000 range, before leveling off with a smaller increase into 2024. Grievances follow a similar trend, with a strong early rise and more gradual growth in later years. For decisionmakers, this pattern signals that while claim denials and appeals increases rapidly, the rate of growth could be and should be slowing down, which could reflect policy changes, market stabilization, or changes in consumer behavior.

The dashboard supports this analysis through interactive filtering by insurer, allowing users to isolate and compare insurer's trends. Additionally, year-over-year comparisons help decisionmakers detect sudden spikes, which may indicate recent policy changes or operational shifts that require review. The accessibility and interactive features allow users to move beyond static summaries and actively explore where and why these changes are occurring.

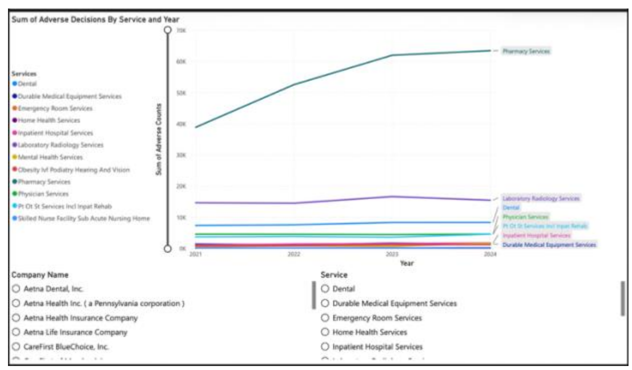

Figure 5. Trends by service (adverse decisions)

Figure 5 focuses on adverse decisions by service over time, another key request from Commissioner Marie Grant was to better understand how denials vary across different types of services. The data reveals a clear and important pattern that pharmacy services is a major outlier. They consistently account for significantly higher adverse decision counts compared to the other service categories and show a strong upward trend across the years. For decisionmakers, this highlights pharmacy services as an area of concern , suggesting that policies related to medications, such as formulary restrictions or prior authorization requirements, may be driving a large portion of denials.

The dashboard enhances this insight through interactive filtering by both insurer and service, allowing users to select specific areas to view. For example, decisionmakers can isolate pharmacy trends for a company or compare how multiple services behave over time for that company. The hover-over accessibility feature further strengthens the analysis by displaying exact values for each service for that year, making it easier to interpret changes without overcrowding the visual. Together, these features turn the visualization into a practical tool for identifying where intervention may be most needed.

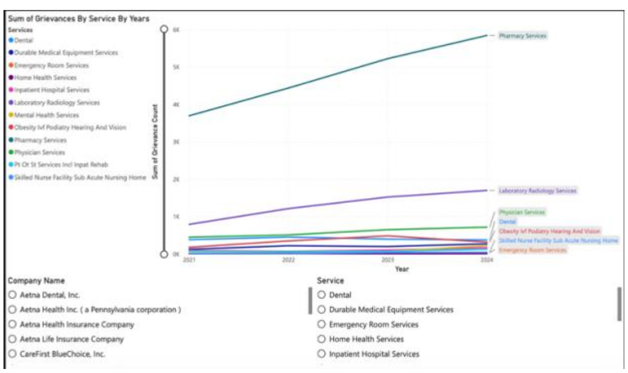

Figure 6. Trends by service (grievances)

Figure 6 shows the number of grievances by service, showing the focus on service trends. The data closely mirrors the adverse decision patterns, with pharmacy services again standing out as the highest category. This reinforces the idea that pharmacy denials are not only frequent but also more likely to be challenged by consumers. For decisionmakers, the high denial rates and high grievance rates signals a potential gap between insurer decisions and consumer expectations or needs.

At the same time, the data reveals important differences across services. Some categories show high adverse decision counts but lower grievance rates. For example, dental services rank among the third highest in adverse decisions but forth in grievances underneath physician services. This suggests that consumers may be more likely to challenge certain types of denials over others, possibly due to cost or urgency. These insights help decisionmakers prioritize areas where consumer dissatisfaction is most pronounced. The dashboard's hover-over accessibility feature allows users to view exact values for each service, making it easier to compare these differences and better understand where disputes are most concentrated.

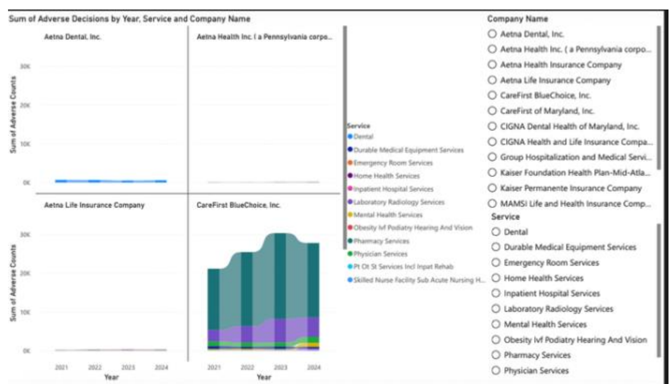

Figure 7. Insurer comparison for adverse decisions (ribbon or multi-panel chart)

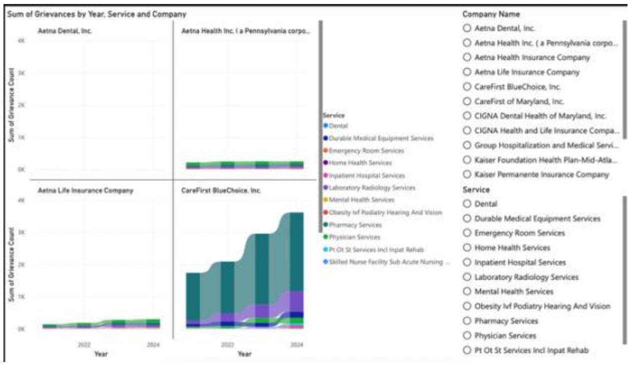

Figure 8. Insurer comparison for grievances (ribbon chart)

Figure 7 and 8 use ribbon charts to compare insurers across different services and years for adverse decisions and grievances. These visualizations emphasize not just volume, but also how the ranking of services changes over time within each company. The movement of the ribbons provides a clear, visual story of a shift in priorities and problem areas. For example, for United Health Insurance Company, pharmacy services had the highest number of adverse decisions in 2021, but in 2022 laboratory and radiology services ranked first. This change in ranking is immediately visible through the repositioning of the ribbons.

For decisionmakers, this type of visualization is especially valuable because it highlights dynamic changes that may not be obvious in static charts. It allows them to track how issues change within a company and identify emerging problem areas before they become widespread. This dashboard also has hover-over accessibility feature that shows the differences in number between two years for a service and the rank of that service. This supports more proactive and targeted decision-making.

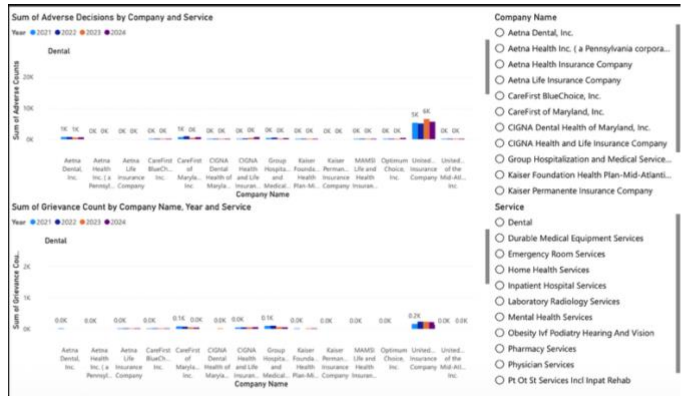

Figure 9. Comparison of Adverse Decisions and Grievances Across Companies and Services Over Time

Figure 9 uses clustered column charts to compare adverse decisions and grievances across companies and services over time. This visualization is designed for direct comparison, making it easier to evaluate how companies perform with one another within the same service category. Each group of bars represents a company, while the colored bars within each group represents a year. This structure allows decisionmakers to quickly identify which companies consistently report higher volumes and whether those trends are increasing or decreasing over time.

The data tells a story that is essential for accountability. Decisionmakers can use this view to identify whether increases are industry wide or only within specific companies. If one insurer shows a disproportionate rise in adverse decisions, it provides a starting point for further investigation into internal policies or practices. By presenting multiple data points such as company, service, and time in a single visualization, the dashboard makes complex comparisons more accessible and actionable.

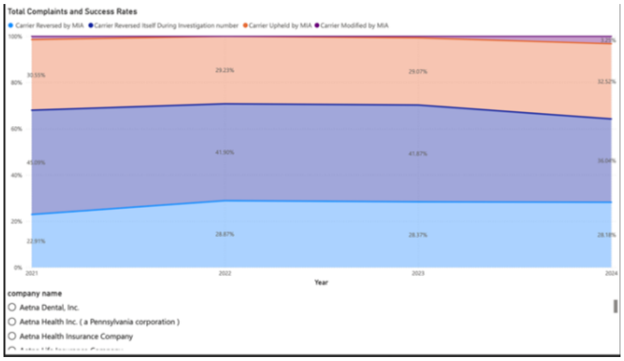

Figure 10. Complaints outcome

Figure 10 is a 100% stacked chart that presents complaint outcomes as percentages, including reversed (treatment deemed necessary), upheld (treatment deemed unnecessary), and modified (adjusted decision). Commissioner Marie Grant requested insight into complaint outcomes to better understand the system's success rate, which is the rate of complaints that are overturned. The data reveals that a significant portion of complaints are reversed, either by the Maryland Insurance Administration or by insurance carriers during the investigation process.

For decisionmakers, this finding is critical. A high reversal rate may indicate that initial claim decisions are not always aligned with final determinations, pointing to inefficiencies or flaws in the decision-making process. The dashboard supports this analysis with data labels that display percentages, as well as a hover-over accessibility feature that provides exact counts alongside those percentages. This allows users to interpret both relative and absolute. Over time, tracking changes in these outcome distributions will be important. For example, if the reversal rate begins to decline and more complaints are upheld, it may prompt a review of whether the system is becoming more restrictive or if earlier decisions are improving in accuracy.

Conclusion:

In conclusion, the dashboards and analysis provide a clear, data-driven view of how adverse decisions and grievances have evolved over time, highlighting key areas such as the impact of pharmacy services and the differences in how consumers respond to different types of denials. By combining interactive features with accessible visualizations, this project allows decisionmakers to move beyond surface level trends and identify where deeper investigation or policy changes may be needed.

While the findings are meaningful, the analysis is limited to a four-year period and may not fully capture long-term trends. Future work could expand the dataset or include additional variables to better understand the causes behind these patterns. Overall, this project demonstrates how data can be used to analyze complex systems and provide insights that can help improve outcomes for consumers.

Acknowledgements:

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